
Complements of Operational Research

Exercise sheet 6 — Queueing

The exercises marked **homework** are to be solved at home and sent by email to the lecturer, by the deadline given in class; the solution is shared afterwards. The others are from previous tests and exams. Unless stated otherwise, Poisson arrivals, exponential service times and FIFO discipline; the waiting cost applies to the customers in the queue (L_q).

Exercise 1 — How many supermarket checkouts to open (I)

Homework

A supermarket needs to decide how many checkouts should be open. On average, 18 customers arrive per hour and a customer is served, on average, in 4 minutes; exponential arrivals and service ($M/M/s/GD/\infty/\infty$). The hourly wage cost is 20 m.u. and the cost per waiting customer is 0.25 m.u. per minute.

- How many checkouts should be open to minimise the total cost?
- How many checkouts ensure that the probability of a customer having to wait in the queue is less than 5%? What is the total cost in that configuration?

Exercise 2 — Bank with two tellers

Homework (exams, 10/07/2018 and 26/07/2019)

Consider a bank with 2 tellers open. On average, 80 customers per hour arrive at the bank and wait to be served. Each teller takes, on average, 1.2 minutes to serve a customer ($M/M/2$).

- The expected number of customers in the bank's queue.
- The mean time spent by a customer in the bank.
- For what fraction of the time is each teller idle? And both at the same time?

Exercise 3 — How many supermarket checkouts to open (II)

Homework (2nd test, 11/06/2024)

A supermarket needs to decide how many checkouts should be open. On average, 32 customers arrive per hour and a customer is served, on average, in 4 minutes; exponential arrivals and service ($M/M/s/GD/\infty/\infty$). The hourly wage cost is 20 m.u. and the cost per waiting customer is 0.24 m.u. per minute.

- How many checkouts should be open to minimise the total cost?
- How many checkouts ensure that the probability of a customer having to wait in the queue is less than 5%? What is the total cost in that configuration?

Exercise 4 — Technicians for a plastic injection-moulding section

Homework (new)

A factory has 6 plastic injection-moulding machines. Each machine breaks down, on average, once every 8 hours of operation; a repair takes, on average, 1 hour; everything exponential. Each technician costs 25 €/h and each idle machine costs 80 €/h.

- With 1 technician and with 2 technicians, compute the mean number of idle machines, the mean number waiting for repair and the mean time a broken-down machine stays idle.
- How many technicians should be hired?
- A colleague used the $M/M/1$ model with $\lambda = 6 \times 0.125$ breakdowns/h. What result did they get for the number of idle machines with 1 technician? Comment.

Exercise 5 — Photocopying service

2nd test, 01/06/2018

At a photocopying service with a single working machine, on average 5 customers arrive per hour and the clerk can serve, on average, 6 customers per hour (M/M/1).

- a) Mean number of customers in the system (waiting or being served).
- b) Time a customer spends in the system.
- c) Mean number of customers in the queue.
- d) Mean time a customer waits in the queue.
- e) Percentage utilisation of the server.
- f) Probability that the clerk is idle.
- g) Probability that there are 2 customers in the system.

Exercise 6 — Doctor's surgery

2nd test and exam, 27/06/2019

A doctor spends, on average, 20 minutes with each patient and the mean time between patient arrivals is 30 minutes (M/M/1).

- a) The arrival rate and the service rate.
- b) Mean number of customers in the system.
- c) Mean time a customer spends in the system.
- d) Mean number of customers in the queue.
- e) Mean time a customer waits in the queue.
- f) Probability that the surgery is empty.
- g) Probability that there are 2 customers in the surgery.

Exercise 7 — Establishment with limited capacity

exam, 05/07/2021

An establishment receives, on average, 450 customers per hour and, on average, a customer is served every 6 seconds. Solve with the M/M/1/GD/K/∞ model with limit $K = 300$; exponential arrival and service times.

- a) What is the mean queue length (number of customers)?
- b) What is the probability that there is a queue (in %)?
- c) What is the mean time a customer spends in the system (in seconds)?

Exercise 8 — How many supermarket checkouts to open (III)

exam, 02/07/2024

A supermarket needs to decide how many checkouts should be open. On average, 14 customers arrive per hour and a customer is served, on average, in 5 minutes; exponential arrivals and service (M/M/s/GD/∞/∞). The hourly wage cost is 25 m.u. and the cost per waiting customer is 0.26 m.u. per minute.

- a) How many checkouts should be open to minimise the total cost?
- b) How many checkouts ensure that the probability of a customer having to wait in the queue is less than 5%? What is the total cost in that configuration?